

2. Week 2 – R Basic Operations

R Basic Operations 1

Question 1

Create a code to show the invoice message to the customer using the appropriate thousands separator. Assumed the customer's name is "Jill", the item is "iPhone 16 Plus 256", the item price is \$1,675.4, the tax is 10%, and the expected outcome should be

"Dear Jill, your total invoice is for iPhone 16 Plus 256 including tax is \$1,842.94". The value should be taken through user input. Hint: Use `readline()` function. Make sure to change the data type accordingly.

Try to use several concatenation strategies and see what you get. Do they produce identical

```
{r}
# Take user inputs
cust_name <- readline("Enter customer name: ")           # Customer's name
item <- readline("Enter item name: ")                   # Item name
currency <- readline("Enter currency symbol: ")         # Currency symbol
itemprice <- as.numeric(readline("Enter item price: ")) # Convert to numeric
tax <- as.numeric(readline("Enter tax percentage: "))    # Convert to numeric

# Calculate the total price with tax
total_price <- itemprice + (itemprice * tax / 100)

# Format total price to include thousands separator and 2 decimal places
formatted_price <- format(total_price, big.mark = ",", nsmall = 2)

# Using paste() for concatenation
msg_paste <- paste("Dear", cust_name,
                  ", your total invoice for", item,
                  "including tax is", currency, formatted_price)

# Using sprintf() for formatted output
msg_sprintf <- sprintf("Dear %s, your total invoice for %s including tax is %s",
                      cust_name, item, currency, formatted_price)

# Using cat() for output
cat("Dear", cust_name,
    ", your total invoice for", item,
    "including tax is", currency, formatted_price, "\n")

# Print the messages using different methods
print(msg_paste) # paste() output
#print(msg_sprintf) # sprintf() output
```

R Basic Operations 2 (Logical Control)

Question 1

Write a function in R named `is_prime` that checks whether a given number is a **prime number** or not. A prime number is a number greater than 1 that has no divisors other than 1 and itself.

Tasks:

1. Define the function `is_prime(n)` that:
 - Returns TRUE if `n` is a prime number.
 - Returns FALSE otherwise.
2. Use a for loop inside the function to check divisors from 2 to $\sqrt{n}$.
3. Get the number from user input
4. Test the function for the numbers 7, 10, and 13 using an if statement to print messages like:
 - "7 is a prime number."
 - "10 is not a prime number."
 - "13 is a prime number."

```
{r}
# Define the is_prime function
is_prime <- function(n) {
  if (n <= 1) { # Prime numbers are greater than 1
    return(FALSE)
  }

  # Loop to check divisors from 2 to sqrt(n)
  for (i in 2:sqrt(n)) {
    if (n %% i == 0) { # If divisible, not a prime
      return(FALSE)
    }
  }

  return(TRUE) # If no divisor found, it's a prime number
}

# Test the function
numbers <- as.integer(readline("Give me a number: "))
if(is_prime(numbers))
{
  msg<-paste(as.character(numbers),"is a prime number", sep=" ")
} else
{
  msg<-paste(as.character(numbers),"is not a prime number", sep=" ")
}

print(msg)
```

Question 2

Write a function called `guess_number()` that implements a **number guessing game**. In this game:

1. The program generates a random number between 1 and 100.
Use `sample(1:100,1)` function to get the random number.
2. The user has to guess the number.
3. Use a while **loop** to keep prompting the user until they guess the correct number.
4. Provide hints:
 - a. Print "Too high!" if the guess is greater than the random number.

- b. Print "Too low!" if the guess is smaller than the random number.
5. When the correct number is guessed, print "Congratulations! You guessed the number." and stop the loop.

Extra Challenge:

Count the number of attempts it took for the user to guess correctly.

```
{r}
# Define the function for the guessing game
guess_number <- function() {
  # Generate a random number between 1 and 100
  target <- sample(1:100, 1)
  attempts <- 0 # Counter for attempts

  # Start the guessing game loop
  repeat {
    # Prompt the user for a guess
    guess <- as.numeric(readline("Guess a number between 1 and 100: "))
    attempts <- attempts + 1 # Increment the attempts

    # Check the guess and provide feedback
    if (is.na(guess)) {
      cat("Invalid input. Please enter a number.\n")
    } else if (guess > target) {
      cat("Too high!\n")
    } else if (guess < target) {
      cat("Too low!\n")
    } else {
      cat("Congratulations! You guessed the number in", attempts, "attempts.\n")
      break # Exit the loop
    }
  }
}

# Run the game
guess_number()
```